

Review

Membrane-based technologies for biohydrogen production: A review

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ABSTRACT

Overcoming the existing environmental issues and the gradual depletion of energy sources is a priority at global level, biohydrogen can provide a sustainable and reliable energy reserve. However, the process instability and low biohydrogen yields are still hindering the adoption of biohydrogen production plants at industrial scale. In this context, membrane-based biohydrogen production technologies, and in particular fermentative membrane bioreactors (MBRs) and microbial electrolysis cells (MECs), as well as downstream membrane-based technologies such as electrodialysis (ED), are suitable options to achieve high-rate biohydrogen production. We have shed the light on the research efforts towards the development of membrane-based technologies for biohydrogen production from organic waste, with special emphasis to the reactor design and materials. Besides, techno-economic analyses have been traced to ensure the suitability of such technologies in bio-H₂ production. Operation parameters such as pH, temperature and organic loading rate affect the performance of MBRs. MEC and ED technologies also are highly affected by the chemistry of the membrane used and anode material as well as the operation parameters. The limitations and future directions for application of membrane-based biohydrogen production technologies have been individuated. At the end, this review helps in the critical understanding of deploying membrane-based technologies for biohydrogen production, thereby encouraging future outcomes for a sustainable biohydrogen economy.

1. Introduction

1.1. State of the art

Reserves of fossil fuels are finite and non-renewable due to the geological timescales on which they are formed. Hence, replacing fossil fuels with renewable and sustainable resources is necessary to safeguard against resource scarcity and prevent environmental effects in future (Chawla et al., 2018; Tawfik et al., 2021b; Hassan et al., 2022). Hydrogen is one of the most promising renewable fuel alternatives owing to several positive features such as a high conversion efficiency to working power (i.e. 99% in fuel cell systems), high energy density, and the absence of greenhouse gas emissions upon combustion (Gassanova et al., 2006; Ghauri et al., 2011; Nikolaidis and Poullikkas, 2017). Hydrogen is produced either by thermochemical or biological routes (Tawfik et al., 2021a, 2021c). The thermochemical routes, including steam methane reforming (76%) and coal gasification (22%) account for

more than 98% of the total hydrogen gas market (Suleman et al., 2015). Nevertheless, these routes are energy intensive, depend on non-renewable resources (fossil fuel) and are environmentally unfriendly (9 kg of CO₂/kg of H₂ produced) (Suleman et al., 2015). On the other side, renewable hydrogen production could come from water electrolysis and pyrolysis of organic waste materials.

Thermochemical and biochemical processes are the main routes for biohydrogen production from organic waste materials. Thermochemical methods include pyrolysis, gasification and supercritical water gasification (Kalinci et al., 2009). Pyrolysis is the process where the substrate is heated in absence of oxygen (Zhao et al., 2019), while, in gasification the substrate is subjected to high temperatures under controlled oxidizing conditions (Robaina et al., 2021). Pyrolysis process produces syngas, biochar and bio-oil, whereas the traditional gasification process produces only syngas (Mavukwana et al., 2021). Both processes are considered energy intensive and environmentally unfriendly (Das and Veziroğlu, 2001). Recently, different biotechnologies for production of

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hydrogen from different sources have been comprehensively inspected, including photodecomposition of organic compounds by photosynthetic bacteria (photo-fermentation), biological-assisted photolysis of water by algae and cyanobacteria, and fermentative hydrogen production from organic waste streams (Hallenbeck and Benemann, 2002; Show et al., 2011). Among these, biological hydrogen production is considered the most economical and lowest energy-demanding process (Da Silva Veras et al., 2017; Hassan et al., 2021b). The main weakness of fermentation, however, is an inefficient substrate conversion rate that results in low hydrogen yields and production rates. Recently, several methodologies have been proposed to tackle this issue, such as optimizing fermentation conditions and bioreactor design, as well as the assembly of more robust and efficient hydrogen-producing microorganisms (Guo et al., 2010; Jung et al., 2011; Boodhun et al., 2017).

Dark fermentation is a bioprocess used to produce hydrogen gas as well as valuable byproducts (e.g. organic acids and alcohols) from organic waste/wastewater without a light source using fermentative bacteria (Tao et al., 2007; Khan et al., 2016; Kumar et al., 2017). It is the most advanced biological hydrogen production technology, and remarkable progress is ongoing toward its practical implementation (Hawkes et al., 2007; Das and Veziroglu, 2008).

The classical continuous stirred tank reactors (CSTRs) have been widely adopted for dark fermentation (Hawkes et al., 2007). CSTRs offer simple construction design, ease of operation, effective homogeneous mixing and operation under varied substrate conditions (Kapdan and Kargi, 2006; Wang and Wan, 2009). Anaerobic membrane bioreactor (AnMBR) that combine fermentation with membrane technology is a promising alternative to achieve high rate biohydrogen production. Compared with CSTR and AnMBR prevents biomass washout from the reactor increasing the solid retention time required for efficient treatment at short hydraulic retention time (HRTs). Furthermore, AnMBRs produce a high-quality effluent and have a smaller plant footprint than CSTRs (Aslam et al., 2014; Charfi et al., 2017). However, membrane fouling remains the main hurdle towards full-scale AnMBR applications.

The beauty of AnMBRs is that they can prevent the accumulation of soluble products such as carboxylic acids and alcohols that can inhibit fermentative hydrogen production, at the same time increasing biohydrogen yields, volatile fatty acid (VFA) production and substrate utilization rates (Jones et al., 2017).

Electrodialysis (ED) is another potential membrane-based technology recently proposed in combination with dark fermentation to further augment hydrogen yields by continuously removing inhibitory volatile fatty acids (VFA) (Hassan et al., 2021a). In ED, an electrical field is applied to initiate the transfer of ions across a selective ion exchange membrane (IEM) from a dilute stream to a concentrated stream (Tao et al., 2016). In fact, ED is receiving attention as a sustainable option for VFA recovery from organic waste (Bak et al., 2019; Hassan et al., 2019). Another emerging membrane-based technology for biological hydrogen production is the microbial electrolysis cell (MEC) (Huang et al., 2011). In MEC, organic substrates are oxidized in the anodic chamber. The electrons are then transferred to the cathode where they combine with protons migrating through a proton exchange membrane to produce molecular hydrogen (Kadier et al., 2016). All these three technologies (AnMBR, ED and MEC) will be discussed in details through the manuscript.

1.2. Motivation and objective of this work

This review analyses emerging membrane-based technologies for the production and optimization of biohydrogen from organic-rich wastewater, focusing on novel membrane-based techniques for maximizing biohydrogen yield such as ED which has been hereby discussed for the first time in a review article. Construction materials of MEC and operation conditions of MBRs targeting maximum hydrogen production have been also surveyed. Literature studies are thoroughly analyzed and compared. In addition, future directions are delineated aiming to

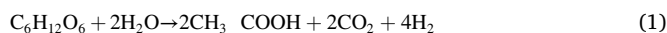
maximize the hydrogen yield in bioreactors and pointing new approaches of membrane applications in bioenergy production.

2. Literature review on membrane-based technologies for biohydrogen production

2.1. Anaerobic membrane bioreactor (AnMBR) for dark fermentative biohydrogen production

2.1.1. Fundamentals of dark fermentation

In dark fermentation, biohydrogen is produced by fermentative microorganism exploiting the electrons produced from organic compounds oxidation to reduce protons to hydrogen gas (Li and Fang, 2007; Hassan et al., 2020; Tawfik et al., 2021a). Dark fermentation has many advantages over other biological hydrogen production routes including its lack of a need for light, wide feedstock range, high hydrogen production rates and simple reactor design and operation (Wang and Wan, 2009; Hallenbeck et al., 2012; Hassan et al., 2020). Dark fermentative biohydrogen production can follow many pathways depending on the microbial species, operational conditions, and feedstock. In general, sugars are first fermented into pyruvate, resulting in energy production in form of adenosine triphosphate (ATP) and nicotinamide adenine dinucleotide (NADH). Pyruvate is further oxidized into acetyl-CoA through ferredoxin reduction (Fig. 1). In case acetyl-CoA is transformed into acetate (Eq. (1)), both NADH and reduced ferredoxin are used to convert H^+ to H_2 through a metalloenzyme called hydrogenase, yielding the theoretical maximum of 4 mol H_2 /mol hexose sugar (Thauer et al., 1977). However, depending on the microbial community and on operation parameters, several fermentation pathways overlap, resulting in yields typically ranging between 14 and 70% of the theoretical maximum (Saady, 2013). In mixed culture fermentation, hydrogen production is often associated with the butyrate pathway (Eq. (2)).



2.1.2. AnMBR configuration

AnMBR configurations can be classified as submerged/immersed and external/side stream (Fig. 2) (Aslam et al., 2018). In the first case, membranes are submerged in the liquid phase of the bioreactor however, in the side stream configuration, the liquid filtration membrane unit is linked to the reactor externally in a separate unit.

In the submerged stream configuration (Fig. 2A), the membrane unit is placed inside the reactor, where the treated effluent is collected by filtration through the membrane whilst the biomass is retained. The AnMBR produces a similar effluent quality as the MBR, but avoiding the costs associated with aeration (Baek and Pagilla, 2006) and producing valuable hydrogen and soluble product streams. The trans-membrane pressure (TMP) of the submerged AnMBR stream has been reported in the range 21–103 kPa (3–15 psi) (Skouteris et al., 2012). The bulk cross-flow velocity achievable in submerged membrane systems tends to be less than 0.6 m/s (Bérubé and Lei, 2006). This could be achieved at suspended material ranging from approximately 50 to 500 μ m. The particle size of the suspended matter in the bulk liquid can have a significant effect on the permeate flux. When the suspended matter is smaller than the membrane pores, it may enter the pores causing irreversible fouling. Otherwise, when the suspended particles are bigger than the membrane pores, they accumulate over the membrane surface forming a cake layer (Salahi et al., 2010). The type of membrane used widely affects the process performance. Stuckey and Hu (2003) reported that the permeate flux obtainable in an AnMBR with a hollow-fiber submerged membrane configuration was slightly higher than that of a flat-sheet submerged membrane configuration (20–35 LMH for hollow fiber while 5–10 LMH for flat sheet). This was attributed to the higher

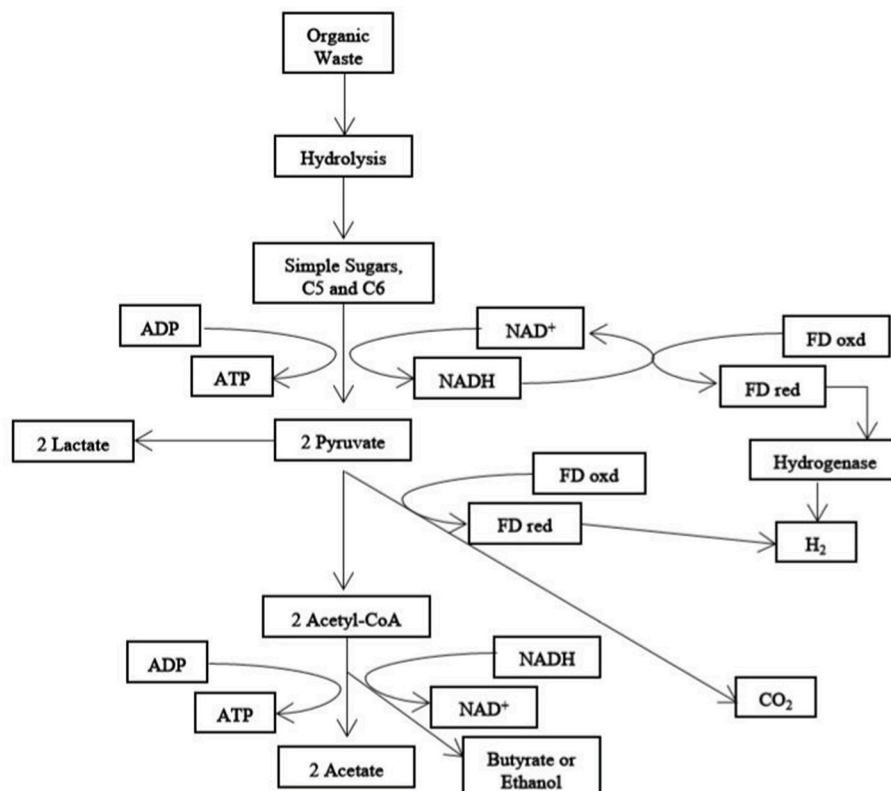


Fig. 1. Hydrogen production pathways in the dark fermentation process; Modified from Hallenbeck et al. (2012).

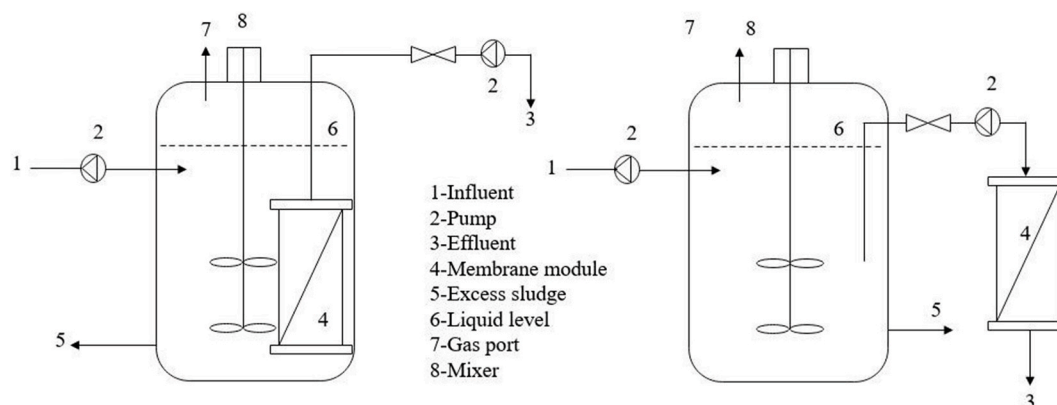


Fig. 2. Schematic representation of AnMBR (A) submerged, (B) external (Aslam et al., 2018).

contact surface of the hollow fibers than the flat-sheet membrane. A similar observation was postulated by Bérubé and Lei (2006), who demonstrated that the physical contact between membranes and reactor broth significantly affects permeate flux.

External configurations (Fig. 2B) of AnMBR includes the recirculation of the mixed liquor through a membrane module outside the bioreactor. Permeate side of the membrane is the collection point for the treated water, and the retained biomass is reverted to the bioreactor. Both rate of circulation (*i.e.*, cross flow velocity through the membrane system) and operating TMP used in external systems are typically high with values ranging from 1 to 5 m/s and 207–690 kPa, respectively.

2.1.3. AnMBR construction materials

Membrane material plays an important role in the operation/maintenance of membrane-based systems used for biogas production, with fouling being a major bottleneck for the adoption of such technology at

large scale. Polymeric and ceramic membranes are common material classes used in membrane fabrication. The most common polymers used for fabricating polymeric membranes are polyvinylidene fluoride, polyethersulfone, polyethylene, polypropylene, and polysulfone (Abdel-Karim et al., 2017). Such polymers require modification to become hydrophilic and suitable for membrane-based applications. Ceramic membranes, as inorganic membrane materials, offer desirable features such as high resistance to thermal and chemical degradation, corrosion, and fouling, which facilitate backwashing and improve their durability over polymeric membranes. However, the high cost of ceramic membranes is hampering their use in MBR systems (Judd, 2016).

2.1.4. Factors affecting dark fermentation processes for biohydrogen production in AnMBR

There are several factors affecting biohydrogen production such as temperature, pH, undissociated volatile fatty acid concentrations,

hydrogen partial pressure (P_{H_2}), HRT and organic loading rate (OLR). The effect of each factor is discussed in detail in the next sections, focusing on studies on fermentation in AnMBRs, summarized in Table 1.

2.1.4.1. Temperature. Temperature is one of the key parameters affecting biological processes as it can shape the microbial community, metabolic activity and metabolic products by enhancing or inhibiting the growth of different species (Karadag and Puhakka, 2010). In addition, kinetics and thermodynamics of the biological reaction depend on temperature (Verhaart et al., 2010). Thermophilic fermentation processes have faster reaction kinetics resulting in faster hydrolysis and biogas production (Dessi et al., 2017). In addition, the inhibition of some hydrogen-consuming bacteria such as homoacetogens has been reported at high temperatures (Luo et al., 2011). Furthermore, high temperature decreases the hydrogen solubility in the liquid phase and consequently decreases the P_{H_2} leading to higher hydrogen yield (Lee et al., 2010). Dessi et al. (2017) obtained a maximum yield of 1.85 mol H_2 /mol xylose at 55 °C compared to 1.2 mol H_2 /mol xylose at 37 °C. However, mesophilic fermentation is often preferred due to the lower energy requirements for heating (Ma and Su, 2019).

2.1.4.2. pH. Operational pH is the most important factor affecting the hydrogen yield (El-Qelish et al., 2020). Fermentation has been reported in a wide pH range (3.0–12.5) (Akroum-Amrouche et al., 2011). While slightly acidic pH (<6) is favourable for microorganisms metabolizing carbohydrate-rich substrates (Akkerman et al., 2002) and can inhibit competing methanogenic (Fernandes et al., 2010), alkaline pH is favoured for protein-based substrates. Indeed, high pH enhances the hydrolysis of proteins, which are for example abundant in sewage sludge, and the activity of alkalotrophic microorganisms producing hydrogen from the resulting amino acids (Lin et al., 2006). Wan et al. (2016) reported homoacetogenesis inhibition at thermophilic anaerobic fermentation of sewage sludge at constant pH 10. Neutral pH is commonly avoided as it enhances the activity of the methanogenic microorganisms which consume the produced hydrogen. Lin et al. (2011b) and Hassan et al. (2018) reported an optimal pH of 5.5 for mixed culture hydrogen production from sucrose (1.04 mol H_2 /mol sucrose) and food waste (0.1 mmol H_2 /g COD removed), respectively. This was supported by a study on fermentation of sugar and starch which gave the highest hydrogen yield at pH in the range 5.5–5.7 (Khanal et al., 2004). The pH also controls the metabolic pathway and the produced metabolites (Wang et al., 2006). Dessi et al. (2017) reported a switch of the metabolic pathway from acidogenesis to solventogenesis (and H_2 consumption) when the pH decreased below 4.5.

2.1.4.3. Hydrogen partial pressure (P_{H_2}). High P_{H_2} inhibits further hydrogen production and makes hydrogen-consuming reactions more thermodynamically favourable (Guo et al., 2010; Oh et al., 2009). When

the hydrogen concentration in the liquid phase is high, the reduction of protons to hydrogen is thermodynamically unfavourable (Mizuno et al., 2000). Many strategies have been applied for P_{H_2} reduction, such as nitrogen sparging (Mizuno et al., 2000), continuous gas release (Logan et al., 2002), larger headspace (Oh et al., 2009) and vacuum stripping (Domenico Foglia et al., 2011). An increase in the hydrogen yield by 78% and 56% was reported with nitrogen sparging when glucose and xylose were used as substrates (Nguyen et al., 2010). Recently, studies on the application of membrane separation have been implemented for continuous extraction of the produced hydrogen, obtaining both low P_{H_2} and high-grade hydrogen gas (Aslam et al., 2018).

2.1.4.4. Hydraulic retention time (HRT). The optimal HRT depends on several factors including reaction rates (function of the substrate type and concentration), active biomass concentration and composition, and temperature. Biohydrogen production from non-sterile waste materials contains a variety of microorganisms that, besides H_2 producers, include substrate competitors and hydrogen consumers (Kim et al., 2004). Methanogens have been identified as one of the main consumers of produced hydrogen. Low HRT can be implemented to limit the growth of methanogens (Fang and Liu, 2002). However, low HRT in continuous flow reactors can also limit the overall substrate utilization (Oh et al., 2009; Hassan et al., 2018).

In dark fermentation, the optimal HRT is usually between a few hours to one day (Yang et al., 2006; Scoma et al., 2013). During a simulation of an integrated biohydrogen fermentation system, it was observed that a short HRT and less OLR favour the process (Kaparaju et al., 2009). When the HRT was increased from four to 8 h, the hydrogen yield obtained from fermentation of common reed by a mixed culture increased up to a maximum of 144.35 mL H_2 /COD removed. However, the H_2 yield decreased when the HRT was further increased to 12 h (106.23 mL H_2 /g COD removed) (Tran et al., 2021). In general, the use of a high substrate to sludge ratio increases the required HRT, whereas high process temperatures encourage reaction kinetics resulting in lower optimal HRT.

2.1.4.5. Organic loading rate (OLR). The OLR refers to the amount of organic material per unit of reactor volume which is subjected to fermentation in each time unit. The optimal OLR of a fermentative bioreactor depends on several factors, including sludge loading rate, pH, substrate type and concentration and reactor design. Fermentative biohydrogen production causes a pH decrease due to the production of VFAs that inhibit further substrate conversion (Mariakakis et al., 2011). This limits the application of high OLR in biohydrogen fermentation. Studies using sucrose in a pilot scale indicated that the highest hydrogen productivity (1.74 mol H_2 /mol sucrose) and the highest percentage (35.3%) of hydrogen in biogas occurred at an OLR of 120 g COD/(L·d) (Lin et al., 2011a). Higher OLR (240 gCOD/(L·d) (Ma and Su, 2019)

Table 1

Summary of studies on biohydrogen production from different substrates in AnMBRs operated at different pH, temperature, and HRT.

	AnMBR module ^a	Carbon source	T (°C)	pH	HRT (h)	R % ^c	H ₂ yield	Ref.
1	PF-MF	Glucose	37	5.5	92.0	–	1.13 mol H_2 /mol glucose	Bakonyi et al. (2015)
2	HF	Glucose	23	5.5	8.0	–	0.008 mol H_2 /gCOD	Shen et al. (2010)
3	CF	Glucose	– ^b	5.5	12	–	2.86 mol H_2 /gCOD	Oh et al. (2004)
4	HF-UF	Glucose	23	5.5	8.0	–	1.49 mol H_2 /mol glucose	Shen et al. (2009)
5	HF	Glucose	37	5.5	4.69	95	1.1 mol H_2 /mol glucose	Li and Fang (2007)
6	HF-MF	Glucose	35	6	–	–	1.1 mol H_2 /mol glucose	Trad et al. (2015)
7	DMBR	Glucose	25	5.5	3.0	100	1.78 mol H_2 /mol glucose	Sim et al. (2020)
8	SMBR	Glucose	60	5.5	–	90	2.87 mol H_2 /mol glucose	Engliman et al. (2020)
9	CMF	Cellobiose	60	7.0	Batch	–	0.68 mol H_2 /mol hexose	Singer et al. (2018)
10	CMF	Avicel	60	7.0	Batch	–	1.21 mol H_2 /mol hexose	Singer et al. (2018)

^a PF-MF, plate-frame type microfiltration; HF, hollow-fiber membrane; CF, cross-flow membrane; HF-UF, hollow-fiber ultrafiltration module; HF-MF, hollow-fiber microfiltration module; DMBR, dynamic membrane bioreactor; SMBR, submerged attached MBR; CMF, coiled microfiltration tube membrane.

^b Temperature has not been specified.

^c Organic removal efficiency.

caused a decrease of the hydrogen yield (1.04 mol H₂/mol sucrose) (Van Ginkel and Logan, 2005; Chen et al., 2008). This argument has been supported by studies indicating that the maximum hydrogen yield is achieved at a lower OLR than that required to achieve the maximum hydrogen productivity (Han et al., 2012). It is thus important that both high hydrogen productivity and overall hydrogen yield are considered when determining the optimal OLR.

2.2. Pilot scale systems for fermentative biohydrogen production

2.2.1. Current status

Fermentative AnMBRs or other bioreactors such as CSTR, anaerobic baffled reactor (ABR) and upflow anaerobic sludge blanket (UASB) are widely investigated at lab scale because of the simple operation and control (Table 2). However, larger-scale studies in pilot or full scale reactors using real wastewaters are still limited (Lu et al., 2019). Full scale application is associated with some practical complications including sludge washout, process parameter control, poor mixing and homogeneity. Controlling process parameters (pH, temperature, hydrogen partial pressure, F/M ratio, organic loading rate and shear force) is crucial to achieve high process efficiency and to avoid reactor failure (Nasr et al., 2021). CSTR is the most applied bioreactor for biohydrogen production process from different waste streams. Micolucci et al. (2020) produced hydrogen from food waste by a 200 L CSTR at 55 °C, at pH 5.2, and achieved a specific hydrogen production of 74.0 L H₂/kg VS. Lin et al. (2010) applied 400 L pilot-scale stirred system for biohydrogen production at 35 °C from sucrose at OLR of 20 g COD/L and HRT of 12 h. This achieved a stable hydrogen production rate of 2.7 m³/m³. day at pH 5.5–6.0, with dominance of two hydrogen producing bacterial species (*Clostridium butyricum* and *Clostridium pasteurianum*). A 200 days study applied a 1480 L pilot scale continuous flow anaerobic fermenter for hydrogen production from molasses at OLR of 3.11–85.57 kg COD/m³/d, achieved a specific hydrogen production rate of 750 L H₂/kg mLVSS-d (Ren et al., 2006). Another study experimented hydrogen production from food waste in a 500 L pilot scale acidogenic fermentation tank reactor treating food waste at 33 °C, pH 5.2–5.5 and HRT of 21 h produced hydrogen yield of 1.82 mol H₂/mol glucose (Lee and Chung, 2010).

2.2.2. Challenges towards scale up of BioH₂ producing bioreactors from lab to pilot and full scale

The major challenges in scaling up AnMBR for biohydrogen production include two main aspects; one is the optimization of the fermentation process and the other one is membrane fouling (Jabbari et al., 2019). Scaling up the fermentation process will deal with the low hydrogen yield, low stability, and with the optimization of the factors affecting the hydrogen fermentation process including pH, HRT, OLR, P_{H₂}, and temperature (Jarunglumert et al., 2018). Furthermore, more investigation is required to find solutions for membrane fouling. Hereby, the different strategies for fouling control are discussed in view of the full scale scenario.

2.2.2.1. Physical cleaning. This is a simple cleaning method including the backwashing or backflushing of the permeate in a reverse direction of the normal operation for 1–2 min every 8–15 min of operation (Psoch and Schiewer, 2006). This method is useful to tackle the reversible membrane fouling by removing the formed cake of solid attached to the surface of the membrane. Attention should be paid during this process as the membrane may be damaged. Nitrogen or argon could be applied instead of the permeate as the backwashing medium; however continuous application of air backwashing may result in membrane damage (Jabbari et al., 2019). Although physical cleaning by backwashing is considered economically feasible in full scale AnMBR application, it is still the most widely applied technology for full scale submerged AnMBR because of its cost (Le-Clech et al., 2006).

2.2.2.2. Chemical cleaning. This method deals with the irreversible fouling when the physical cleaning is not effective, therefore the application of chemical cleaning is necessary (Wang et al., 2014). Chemical cleaning includes maintenance cleaning, intensive chemical cleaning and chemical backwashing. Chemicals which are used for the chemical cleaning depends on the type of the foulants e.g. sodium hydroxide, sodium hypochlorite for the organic foulants, and organic acids (citric acid and hydrochloric acid) for the inorganic foulants (Meng et al., 2017). This also includes the application of ozone gas and carbon dioxide sparging while the reactors have to be maintained. Although chemical cleaning is an efficient fouling removal method, its frequent application may lead to membrane damage (Aslam et al., 2018). In addition, its application on full scale will consume huge amounts of

Table 2
Pilot scale studies for biohydrogen production from different feedstock.

	Reactor	Reactor volume (m ³)	Feedstock	T (°C)	pH	HRT (h)	H ₂ yield	Reference
1	Anaerobic sequencing batch reactor	0.15	Food waste	35 ± 1	5.3 ± 0.1	36	0.5 mol H ₂ /mol hexose	Kim et al. (2010)
2	Granular sludge bed fermentor	0.4	Sucrose	35	6.0	12	1.04 mol H ₂ /mol sucrose	(Lin et al., 2011a, 2011b)
3	Granular sludge bed fermentor	0.4	Sucrose	35	6.0	12	2.34 mol H ₂ /mol sucrose	Lin et al. (2010)
4	Continuous stirred tank reactor	0.2	Food waste	55	5.2	79.2	3.3 mol H ₂ /g VS	Micolucci et al. (2020)
5	Continuous flow anaerobic fermenter	1.48	Molasses	35	7.0	11.4	1.4 mol H ₂ /g VSS	Ren et al. (2006)
6	Acidogenic fermentation tank reactor	0.5	Food waste	33	5.2–5.5	21	1.82 mol H ₂ /mol glucose	(Lee and Chung, 2010)
7	Bioreactor	3.0	Cane molasses	37	6.5	24	18.74 mol H ₂ /kg COD removed	Balachandrar et al. (2020)
8	Granular sludge bed fermentor	0.4	Sucrose	37	5.6–5.8	8	3.84 mol H ₂ /mol sucrose	(Lin et al., 2011a, 2011b)
9	Continuous stirred tank reactor	100	Sugar cane distillery	Not controlled	Not controlled	48	2.76 mol H ₂ /mol glucose	Vatsala et al. (2008)
10	Anaerobic Baffled reactor	3.0	Sucrose	–	5.57–5.64	24	10.47 ± 0.47 mmol/g	Lu et al. (2018)
11	Up-flow anaerobic sludge blanket fixed-film reactor	0.027	Palm oil mill effluent	37	5.1–5.6	24	1.6 mol H ₂ /day	Akhbari et al. (2021)
12	Continuous flow reactor	3.0	Glucose	35	5.82–6.23	48	3.80 mmol H ₂ /g	Lu et al. (2019)
13	Agitated granular sludge bed reactors	0.4	Condensed molasses	37	6.0	12	1.04 mol H ₂ /mol sucrose	Cheng et al. (2011)

chemical which will increase the running cost of the AnMBR.

2.3. Commercial development of AnMBR technologies

The first application of AnMBR was reported by Grethlein (1978), where an external cross-flow membrane module was applied to treat the effluent of septic tank reactor and achieved an improvement in the removal efficiency of biochemical oxygen demand by 72%. The next three decades have witnessed a lot of improvements in the AnMBR technologies, supported by the investments of the private and governmental sectors. These efforts came up with the first commercial AnMBR by Li et al., (1985), called MARS (membrane anaerobic reactor system). In 1990, another AnMBR called ADUF (anaerobic digestion ultrafiltration) was introduced by Ross et al. (1990). Both MARS and ADUF were applied for industrial wastewater treatment in pilot and full scale. The commercialization of the AnMBR got a great push through a six-year project funded by the Japanese Ministry of International Trade and Industry “Japan’s aqua renaissance ‘90 Project”. This project aimed to find out a low cost wastewater (industrial and municipal) treatment systems coupled with energy recovery, with the application of membrane technologies for water reuse. The project team applied different types of membrane materials (polymeric and ceramic), with different modules (capillary, hollow fibre, tubular and plate), in different configurations including the AnMBR (Kimura, 1991). The project findings indicated that membrane technology is more efficient and consumes one third of the energy required by the conventional treatment plants. In last two decades, the research focused on the operational conditions of the AnMBR including the fouling control, foulants, membrane fabrication and characteristics (Fig. 3). Full scale KUBOTA submerged anaerobic membrane bioreactor (KSAMBR) has been developed in Japan for treatment of food and beverage industries (Kanai et al., 2010). This full scale system aimed to increase the energy efficiency and minimize the technical problems associated with membrane fouling. This system achieved a stable operation process with one third footprint than conventional anaerobic systems, better effluent quality and the produced sludge was 3–5 more concentrated than conventional ones (Dvořák et al., 2016). Therefore, its application was extended to stillage, potato processing and sludge liquor.

From economical and logistic point of view, AnMBR possesses many advantages of the anaerobic (biogas production, and anaerobic degradation) and aerobic (high effluent quality and suspended solids free) treatment technologies. However, the issues related to the operational conditions, capital cost and running cost should be addressed.

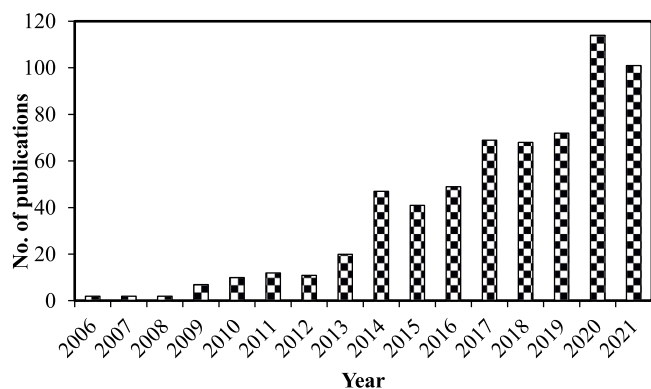


Fig. 3. Number of publication since 2006 onwards according to Scopus using “AnMBR” as the keyword.

2.4. Safety aspects of application of AnMBR

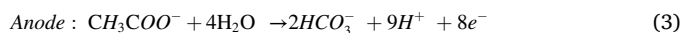
2.4.1. Bacterial contamination

The high quality effluent of the AnMBR opens the way for its direct reuse purposes, therefore, concerns about the possible contamination of the effluent with pathogenic bacterial species or antibiotic-resistant bacterial species should be considered seriously. Membrane based wastewater treatment technologies have shown a significant improvement toward removal of biotic contaminants which allows the reuse of treated wastewater without the need of an extra stage of disinfection (Hai et al., 2014). Besides, the anaerobic digestion process has been reported as effective way for reduction of the burden pathogens during the anaerobic treatment process (Avery et al., 2014). In addition, both membrane technology and anaerobic digestion process have been identified as an effective way to reduce the abundance of the antibiotic resistance genes and antibiotic resistance bacteria (Pruden et al., 2013). AnMBR system couples the membrane technology and the anaerobic digestion process. Furthermore, AnMBR can sustain a high sludge biomass concentration, especially for slow-growing and hydrolytic bacteria and methanogens, which represents an extra advantage and mitigates the possible risk concerns of the sludge application (Li and Yu, 2016). Therefore, the application of AnMBR technology couples many advantages including high effluent quality and possible effluent reuse as its potential role in the reduction of the pathogens and antibiotic resistance biota (Harb and Hong, 2017).

3. Microbial electrolysis cell (MEC)

3.1. Fundamentals of MEC

Developing innovative technologies for hydrogen production will help meeting its growing energy demand. One of these technologies is MEC, which enables hydrogen production from wastewater (Kadier et al., 2016). In MEC, organic substrates are biologically oxidized to CO_2 , electrons and protons in the anodic chamber. The electrons are then transferred to the anode electrode by microorganisms, and travel through a conductive wire to the cathode where they combine with protons migrating through a proton exchange membrane (Fig. 4-A) to produce molecular hydrogen. Membraneless, single chamber MEC (Fig. 4-B) cells have also been proposed to reduce the capital costs. When a MEC cell is fed with acetate as a model substrate, the reactions taking place at the anode and cathode electrode can be summarized as follows (Eqs (3)–(5)):



Hydrogen production does not occur spontaneously in MEC cells. In order to produce hydrogen at the cathode, MEC reactors require an externally supplied voltage typically above 0.4 V (Kadier et al., 2016). The microorganisms have an important role in the MEC process success. During MEC operation, methanogenic microbes compete with electrochemically active bacteria for both VFAs and H_2 substrates. In this way, acetoclastic methanogens convert acetate to methane, while hydrogenotrophic methanogens consume H_2 to produce CH_4 using carbon dioxide as a source of carbon. Therefore, using alternative thinking such as using single pure culture instead of using mixed anaerobic culture may be the best solution achieve higher hydrogen yields with MEC (Kadier et al., 2016).

3.2. Construction materials

3.2.1. Cathode material

The cell cathode is vital for MEC as it hosts hydrogen evolution (Eq.

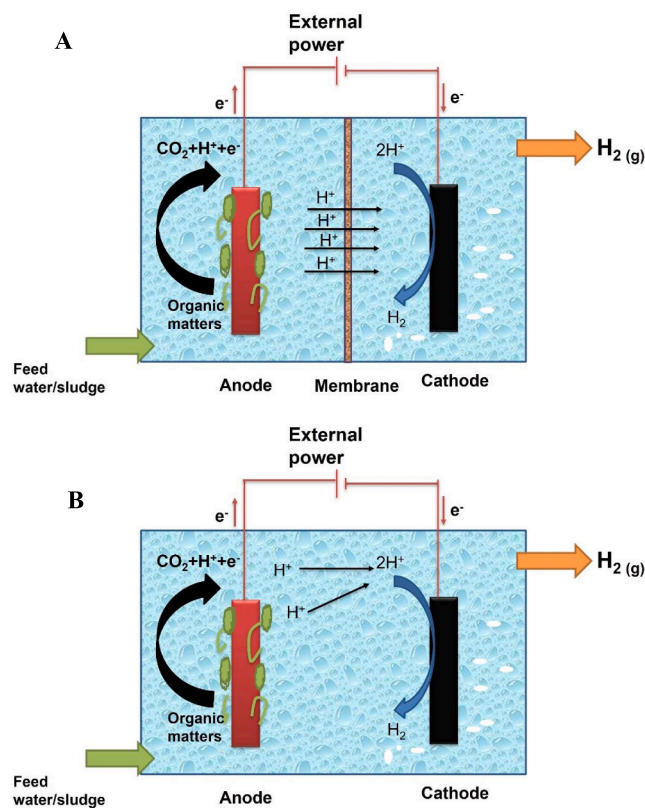


Fig. 4. Schematics of (A) two-chamber and (B) single chamber of MEC (Zhang, 2012).

(4)). Hence, the selection of the cathode material is an important step when deploying MEC technology. Hydrogen production rate and over potential depends on the material type, presence or absence of a catalyst and operation strategy (Morales-Guio et al., 2014). Platinum is an efficient catalyst for hydrogen evolution reaction (HER); however, its use for large-scale applications is costly. Several alternatives to platinum such as nickel-based alloys, stainless steel, woven mesh, and graphite felt have been proposed (Table 3). Recently, the use of carbon-based materials has shown promising results in combination with suitable catalysts (Varanasi et al., 2019). Huang et al. (2011), Dai et al. (2016)

Table 3

Summary of studies on hydrogen production in MECs equipped with different cathode materials.

MEC design	Cathode materials	H ₂ production rate (m ³ /m ³ /day)	Refs.
Single chamber	Carbon	3.2	Manuel et al. (2010)
	Paper	0.63	Dai et al. (2016)
	Carbon paper	1.5	Selembo et al. (2009)
	Stainless steel	1.8	Tenca et al. (2013)
	Stainless steel	3.9	Tartakovsky et al. (2009)
	Graphite felt		
	Graphite fiber	1.9	Lu et al. (2011)
	Ni Foam	5	Jeremiasse et al. (2010)
	Carbon Cloth	0.74	Kiely et al. (2011)
	Carbon Cloth	0.17	Cusick et al. (2010)
Two-chamber	Graphite felt	2.4	Croese et al. (2014)
	Stainless steel wire wool	5×10^{-6}	Heidrich et al. (2013)
	Graphite felt	0.63	Rozendal et al. (2008)

and Manuel et al. (2010) used the same cathode material (carbon paper) along with different catalysts (palladium nanoparticles, nano-Mg(OH)₂/graphene and nickel powder, respectively). The highest hydrogen production rate of 0.63 ± 0.11 m³ H₂/m³/d was obtained using nano-Mg(OH)₂/graphene. Other researchers used stainless steel as cathode material (Selembo et al., 2009; Tokash and Logan, 2011) and reported a maximum hydrogen production rate of 1.5 m³ H₂/m³/d. Tokash and Logan (2011) used molybdenum disulphide (MoS₂) as catalyst, obtaining a current production of 10.3 A/m², with gas compositions of H₂ (92 ± 3%), CH₄ (1 ± 0.8%), and CO₂ (6.7 ± 2.3%), and COD removal of 88 ± 2% when using synthetic wastewater. On the contrary, Selembo et al. (2009) used nickel oxide as a catalyst obtaining a negligible current over a time due to the reduction in chemical stability of the oxides.

3.2.2. Anode materials

In MEC, the microorganisms utilize the anode as electron acceptor to carry out anaerobic respiration. Thus, it is essential that the anode material is conductive, noncorrosive, porous, cheap, and biocompatible. Carbon-based materials such as carbon felt, activated carbon and graphite are often used as anode electrode materials (Xie et al., 2015). Porous electrodes have shown higher current densities due to the higher surface area for bacteria attachment. However, porous electrodes could result in limited mass and charge transfer that could severely affect the growth of bacteria. The microbial community prevailing in the anode also plays an important role in determining the hydrogen yield in MECs. Inhibiting the growth of non-electrogenic bacteria at the anode, particularly methanogenic archaea, is essential to obtain high hydrogen yields at the cathode. Different chemical-based inhibitors have been reported in literature like 2-bromomethanesulfonate (BES) and lumazine (Chae et al., 2010). However, use of such chemicals increases the overall cost of the process.

3.2.3. Membrane materials

The anode and cathode of MECs are often separated by ion-exchange membranes (IEMs) to maintain the purity of hydrogen gas and restrict the anodic bacteria to consume the electrochemically produced hydrogen (Zhang, 2012). The most common membranes used are proton exchange membranes, anion exchange membranes, bipolar membranes, and charge mosaic membranes (Kadier et al., 2016). MEC membranes are dense and can be either heterogeneous or homogeneous. Heterogeneous membranes are made of an ion exchange resin, which is milled to a small grain size and then mixed with a solution of binding polymers. The ratio of resin grains and binding polymer determines the electrical and mechanical properties of the membrane. More binding polymer improves the mechanical strength, but also increases the electrical resistance. MEC membranes require the following features: high-ionic conductivity, high selectivity over counter-ion and co-ions, high ion exchange capacity, and moderate mechanical strength to maintain integrity during long-term operation (Van der Bruggen, 2015).

Apart from H₂ purity, another major advantage of using membranes is that it reduces the crossover of hydrogen and the risk of short circuiting (Varanasi et al., 2019). However, membranes could increase the internal resistance of the system, particularly when fouling occurs, and cause pH imbalance between the two electrodes, leading to lower current densities at a given applied voltage and eventually hindering the HER at the cathode. The construction become more costly, if membranes are employed in the MEC designs. Thus, it is recommended to develop a membrane-free design that could help overcome these restrictions (Varanasi et al., 2019).

3.3. Hydrogen production from wastewater in MEC

A variety of real wastewaters have been used as a fuel sources in MECs. Kiely et al. (2011) used potato processing wastewater as a feed and reported a hydrogen recovery of 83% in terms of columbic

efficiency (CE) with a hydrogen percent of 73% and a hydrogen production rate of $0.74 \text{ m}^3/\text{m}^3/\text{day}$. Wagner et al. (2009) used swine wastewater as a feed and reported a hydrogen yield of $0.9\text{--}1.0 \text{ m}^3/\text{m}^3/\text{day}$ with hydrogen percentage of $58 \pm 1\%$ and CE of $29 \pm 17\%$. Cusick et al. (2010) used winery wastewater as a feed and reported a hydrogen production rate of $0.17 \pm 0.09 \text{ m}^3/\text{m}^3/\text{day}$ with a hydrogen percentage of $70 \pm 8\%$ and a CE of $50 \pm 5\%$. In the same study, the authors reported a hydrogen production rate of $0.28 \pm 0.09 \text{ m}^3/\text{m}^3/\text{day}$ and hydrogen percentage of $70 \pm 1\%$ with a CE of $64 \pm 5\%$ when using domestic wastewater (Heidrich et al., 2013). When treating real wastewater, the COD removal should be taken into account when evaluating the efficiency of the system. For instance, Tenca et al. (2013) used industrial wastewater produced from chemical industry and food processing wastewater as a substrate obtaining a maximum hydrogen production rate of $1.8 \text{ m}^3/\text{m}^3/\text{day}$ and $0.8 \text{ m}^3/\text{m}^3/\text{day}$, respectively. However, the total COD removal efficiency was 70% for the industrial wastewater and only 22.5% for the food processing wastewater.

3.4. Pilot scale systems for fermentative biohydrogen production from MEC

Regarding to MEC systems, most of the MEC bioreactors were fed in continuously mode. Cusick et al. (2011) built the first large MEC reactor with a volume of 1000 L to produce hydrogen from winery wastewater. It was noticed that the biogas contained methane besides hydrogen as well however, the cathode in that case was highly polluted. Following, Heidrich et al. (2014) developed a 100 L two-chamber MEC with successful inhibition of methane content in the biogas by aiding a polymeric membrane. This membrane separated the cathode from the anode and helped in avoiding microbial crossover to the catholyte (Rivera et al., 2018). Furthermore, Leicester et al. (2020) used a pilot-scale of MEC fed with sludge liquor for biohydrogen and he reported that the biohydrogen yield was $0.0155 \pm 0.0076 \text{ m}^3/\text{m}^3/\text{day}$ with a removal of COD $24.5 \pm 1.6\%$ at HRT of 6 h.

3.5. Factors affecting MEC based hydrogen production

3.5.1. pH and temperature

In MEC, the HER and the microbial activity at the bioreactor is pH dependent. The bacteria inside MEC are highly sensitive to pH and so, any small variations affects the microbial community and the performance of the MEC (Merrill and Logan, 2009). Furthermore, many parameters such as media conductivity and ion transfer are associated with pH variations. Many researchers studied the effect of pH on the performance of two-chamber MECs. It was suggested that more hydrogen is produced in case of low pH at the cathode chamber and neutral pH at anode chamber (Kyazze et al., 2010). Higher pH in the chambers effect on the proton accumulation due to the metabolism of electrogenic bacteria can be reduced in the anaerobic fermentation system by using the high level of pH. Temperature also has a significant effect on MEC performance. Change in temperature affects the growth and mass transfer rate of the electrogenic bacteria (Omidi and Sathasivan, 2013; Zhu et al., 2014). Omidi and Sathasivan (2013) stated that temperature is a very important factor effect on the activity of the microbial community and the MEC performance. He operated a two of MECs at a serial of temperatures with using of a feed contain a big portion of acetate. The study tells us that the COD removal rate was 16.8 and 36.2 mg/L/h at 25 and 31°C which confirms the highly effect of the temperature on the MEC performance.

3.5.2. Substrate used in MEC

Using a substrate containing the essential ingredients for feeding the microorganisms in MEC system is essential to achieve the best performance (Kadier et al., 2016). Although 12 mol H_2 can be theoretically produced from 1 mol glucose in MEC, the best yields obtained so far were only $2.5\text{--}3.8 \text{ mol/mol}$ (Escapa et al., 2016). According to Merrill

and Logan (2009), the wastewaters containing a big portion of acetate as substrate gives more higher hydrogen yields (Logan et al., 2008).

3.5.3. Hydraulic retention time (HRT) and organic loading rate (OLR)

In addition to the previous MEC operating affecting factors, the hydrodynamic force and organic loading rate on the MEC system could effect on the performance of the MEC (Ajayi et al., 2010). It has been confirmed that the hydrogen production in an MEC was markedly influenced by HRT, but not significantly affected by the anode biofilm. Another considerations in this subsection, it has been stated that the OLR and HRT can be crucial operational factors under the continuous mode operation of MECs (Gil-Carrera et al., 2013). More hydrogen could be obtained by optimizing these operational factors for instance Gil-Carrera et al. (2013), studied the performance of the MEC at different HRT and OLR. He used three HRTs of 4, 7 and 10 h with OLR of 0.79, 0.32 and 0.23 g/L/d . The hydrogen flow rate were 31.1, 52.7 and 11.5 mL/d . This study confirms of the highly effect of HRT and OLR for biohydrogen production from MEC.

4. Electrodialysis

In-line carboxylic acid extraction can be integrated with dark fermentative reactors to alleviate the inhibitory effect of soluble product accumulation. Several membrane-based technologies are currently available for carboxylic acid extraction and concentration, including liquid-liquid extraction, nanofiltration, pervaporation, forward osmosis and electro-membrane technologies such as electrodialysis and membrane electrolysis, which have been compared and discussed in Dessì et al. (2021). Among them, in-line carboxylic acid extraction by electrodialysis has been shown to enhance H_2 production by a factor of 3.75 (Jeremiasse et al., 2010) which makes its integration to fermentative reactors particularly promising. Hence, this section focuses on electrodialysis and its application to enhance hydrogen production in fermentative reactors.

4.1. Fundamentals of electrodialysis (ED)

Electrodialysis (ED) is a technology in which an electrical field is used to promote ion transfer from one stream to another stream across ion-selective exchange membranes (Hassan et al., 2019). ED is a novel technology for treating different types of wastewaters and has been recently applied in combination with fermentation to alleviate the inhibitory effect of VFA accumulation (Hassan et al., 2019). VFA removal can be achieved by separating the ionic and non-ionic components of the fermentation broth alongside ion exchange membrane upon application of a sufficient voltage (typically $10\text{--}20 \text{ V}$) (Hassan et al., 2020). IEMs are selective towards either cations or anions (Al-Amshawee et al., 2020), resulting in the removal of charged components from mixed waste streams to obtain high-quality products (Hassan et al., 2019; Chalmers Brown et al., 2020). From an operational point of view, ED scope and efficiency depend on the type and configuration of membranes used in the electrodialysis stack (Hassan et al., 2019), as detailed in the following sections.

4.2. Electrodialysis configuration

An ED system is composed of a power supply and an electrodialysis stack (Fig. 5). ED stacks comprise of a series of IEMs (alternating AEM and CEM pairs), electrodes and spacers which create a series of dilute and concentrate chambers. An electrolyte solution is also pumped through the electrodes to supply the electrical conductivity and work as a pH buffer for the concentrated VFA streams. With the application of an electric field across the stack, ions migrate through the IEMs from the feed into the concentrate chambers, reducing the ion concentration in the dilute stream. The arrangement of AEMs and CEMs follows the "Anode-CEM-AEM-CEM-AEM-Cathode" configuration which enables

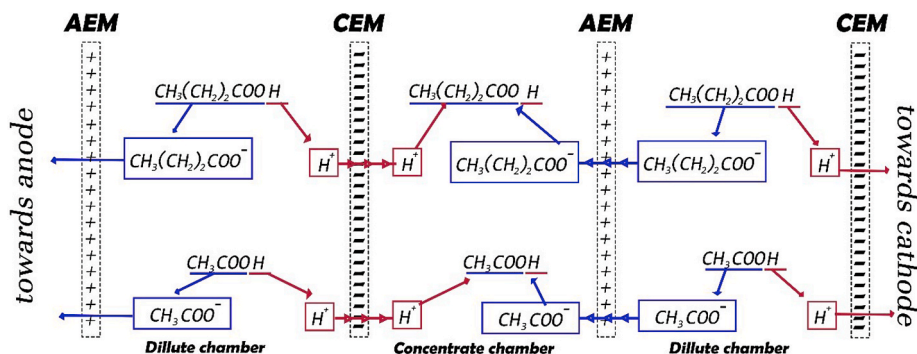


Fig. 5. VFAs recovery by ED technology. Redrawn from ref (Jones et al., 2017).

the simultaneous migration of both cations and anions into the same (concentrate) chamber. The required treatment capacity will determine the scale and extent of the ED stack (Chalmers Brown et al., 2020). At the industrial scale, the cell pairs could be as many as 500 and the membrane area may be as large as 1 m^2 (Al-Amshawee et al., 2020). In laboratory scale units, the active area of a membrane ranges from 0.01 to 0.06 m^2 (Hassan et al., 2019). The distance between membranes in ED stacks may range from 0.3 to 2 mm (Strathmann, 2010). Spacers made of polypropylene or plexiglas materials with an equal area to the effective area of the installed IEMs and 0.42 – 10 mm thickness are installed between the membranes. A fixed voltage is delivered to the system by using a power supply (Luiz et al., 2017). The voltage used in ED system usually ranges between 1 V and 1.5 V for each membrane pair in ED stack (Zhang et al., 2012). Typically, the applied voltage to laboratory scale unit ranges from 7 V to 30 V .

4.3. Integration of ED with fermentation

Several studies reported the application of ED to recover VFAs produced in fermentation reactors. Jones et al. (2017) integrated ED to a sucrose-fed fermenter to alleviate the inhibition effect of VFAs accumulation on hydrogen production. They reported that ED increased the hydrogen production rate by 3.5 times in comparison with the control fermenter without ED. However, the authors could not apply the technology in continuous mode due to the fouling of the membranes inside the ED stack. Such an obstacle has been resolved by Hassan et al. (2019), who for the first time extracted VFAs continuously from a system treating real food waste, increasing the hydrogen yield from $64.7 \text{ mL H}_2/\text{g VS}$ for the control setup to $227.3 \text{ mL H}_2/\text{g VS}$ with the integration of ED. In this study, the fouling issue was minimized by adding a microfiltration (MF) unit as pretreatment before ED to avoid high fouling incidence, and the authors reported only minor fouling issues on the MF unit. However, these experiments were conducted for only 100 h in continuous mode, and thus need to be further investigated for longer times.

Recently, ED efficiency was increased by combining ED membranes with a polytetrafluoroethylene (PTFE) membrane for the removal of VFAs (Chalmers Brown et al., 2020). It was shown that stacking PTFE membranes leads to an increase of the active surface area of the membrane, shortening VFA recovery times. Furthermore, ammonium, phosphate and nitrate remained in the dilution chamber and did not cross the PTFE membrane stack, which represent a unique advantage in fermentation, since allows for the extraction of high-purity carboxylic acids while retaining the nutrients inside the reactors. Up to 98% of total VFA recovery was achieved with the stacked PTFE and ED system. Furthermore, this study confirmed that VFAs could be concentrated 10 times more by coupling two membrane processes (liquid-liquid separation with PTFE membrane and ED) than with ED alone. The process is highly affected by low conductivity in the diluate compartment; however, an increase in the separation efficiency may be achieved with

longer ED operation. The role of pH should be accurately investigated since the separation efficiency can be highly affected by the relative dissociation of each acidic organic species (Chalmers Brown et al., 2020).

5. Perspectives and techno-economic analyses

Hydrogen is expected to play a key role in the low-carbon transition, attracting investments and incentives to fulfil up-to 18% of the global energy demand by 2050 (Yukesh et al., 2021). At present, due to its technological maturity and comparatively low cost, $>90\%$ of the hydrogen is produced by steam reforming of natural gas. However, green production methods are increasingly attracting incentives and investments that could make them competitive in the near future, with an announced production of 11 M tons of clean hydrogen by 2030 (Council, 2021). The political pressure on green energy and on circular economy, is putting organic waste in the spotlight as a cheap, flexible, readily available and widely distributed substrate for hydrogen production by dark fermentation (Alibardi et al., 2020). The abundance of organic raw materials will enable the development of large plants for hydrogen production that will further benefit of the economy of scale (Han et al., 2016).

Despite the intensive research performed in the last two decades, dark fermentation still suffers of low production yields/rates and process instability, resulting in production costs of approximately $2.6 \text{ \$/kg}$, that make it not competitive with steam reforming (cost as low as $1.25 \text{ \$/kg}$) at commercial scale (Park et al., 2021). From a technological perspective, bioreactor concept that ensure high biomass retention, such as AnMBRs, can boost hydrogen production rates by maintaining low hydraulic retention times. Hydrogen production rates as high as $60.5 \text{ L H}_2/\text{L/d}$ have been achieved in AnMBRs at 2 h HRTs (Park et al., 2019), at least twice higher than the highest production rates obtained in CSTRs. Furthermore, the falling costs of hollow fibre membranes, currently in the range of 15 – $25 \text{ \$/m}^2$, will further support the adoption of AnMBR technology for biohydrogen production, although fouling control and long membrane lifespan are crucial factors to avoid excessive operation costs, since membrane replacement costs can be as high as 25% of the total operation costs (Aslam et al., 2018). It should be also considered that full scale AnMBR plants will benefit of the economy of scale, which will result in much lower membrane costs than lab scale reactors, although it was shown that the H_2 production rate correlates negatively with the reactor volume (Qyyum et al., 2022). Hence, the strengths, weaknesses, opportunities and threats (SWOT) of fermentative H_2 production in membrane production in AnMBRs can be summarized as follows (Table 4).

Electrodialysis (ED) is a well established method at industrial scale, which can have significantly lower operation costs than other extraction methods. Energy consumption for the extraction of carboxylic acids by electrodialysis can be estimated as about 5 kWh/kg at typical carboxylic acid concentrations achieved in fermentation (Hernandez et al., 2021),

Table 4
SWOT analysis of fermentative hydrogen production from wastewater in AnMBRs.

Strengths	Weaknesses	Opportunities	Threats
- High production rates - Prevents biomass washout allowing short HRTs - Low reactor footprint - Generation of a clean effluent - Mitigates product inhibition	- Membrane fouling - High installation and operation costs - HPR decreases with reactor size - Requires wastewaters with low solid content - Lower reliability than CSTRs	- Decreasing cost of membranes (both polymeric and ceramic) - Emerging, low-cost dynamic MBRs - Decreasing cost of renewable electric energy - Membrane modules can be added to retro-fit existing plants	- May be outcompeted by simpler reactor configurations - Frequency of backwashing and membrane lifetime dependent on wastewater quality - Non-standardised AnMBR market

although being strictly dependent on the medium conductivity (Lopez and Hestekin, 2015). Despite the potential for preventing product inhibition, the integration of ED in fermentative reactors suffers of major drawbacks. Besides the fouling issues already discussed for AnMBRs, ED suffers of low selectivity, resulting in the competitive extraction of anions other than carboxylates (such as chlorides or phosphates), with the consequent decrease of purity of the extracted products (Scoma et al., 2016). Hence, an extraction process that enables selective extraction of carboxylic acids while maintaining in the fermentative bioreactors nutrients and other ions, such as membrane liquid-liquid extraction, should be applied upstream to ED to achieve the highest product purity (Dessi et al., 2020; Chalmers Brown et al., 2020).

Microbial electrolysis cell (MEC) is a promising, sustainable hydrogen production technology that enables simultaneous wastewater treatment at lower energy input than the traditional activated sludge process. MECs can indeed be powered by renewable electricity sources (e.g. solar or wind power), whose price is expected to decrease at least by 20–30% in the next years (Child et al., 2019). To date, MEC is mainly limited to lab-scale experiments with few pilot scale demonstrations (Baeza et al., 2017). The adoption of MEC at industrial scale is hindered by the high capital cost of the devices. It was estimated that a 90% reduction of the costs of anode electrode, and a lifespan of 20 years, is required to make MEC financially competitive with activated sludge process for treating wastewater (Aiken et al., 2019). However, when considering the whole life cycle, MECs offer a better environmental performance than activated sludge process, having the potential of avoiding emission of as high as 86 tons CO₂/m³ wastewater treated by generating green hydrogen that displaces production from fossil fuels (Zhang et al., 2019). Thus, the increasingly restrictive legislations of carbon emissions, with the resulting carbon credits and green production incentives, can benefit MEC economics towards its commercial viability.

6. Knowledge gaps

Key knowledge gaps that have to be addressed to maximize the benefit of biohydrogen production with membrane technologies as follow:

- 6.1 Storage-related issues, production, transportation and utilization aspects. Critical point is further improvement as well as development needed to improve volumetric and gravimetric density. In addition, transportation and utilization of H₂ fuel requires more knowledge and standards pertaining to safety such as indoor and outdoor operation safety distance, flammability range, leakage sensing, refueling speed control etc. Deploying outreach campaigns to lease such info to public (Nikolaidis and Poullikkas, 2017).

- 6.2 Usage the catalysts is a new window to meet the environmental regulations. Further details regarding this point can be found in this article (Wijayasekera et al., 2022).
- 6.3 Economic evaluation of the integrative pretreatment approaches (Prabakar et al., 2018).
- 6.4 The lack of two-way mutual communication between researchers and engineers is the primary barrier to commercialization because of societal and economic limitations. There is an urgent need to assess the potential outcome at the industrial scale.
- 6.5 Deploying modern environmental-related modelling tools such as density functional theory (DFT), Aspen, and machine learning on the membranes used for biohydrogen production. Basically, these membranes may be micro/ultrafiltration membranes and such membrane kind has been modelled in few number of previous work such as (Abdel-Karim et al., 2020; El-Naggar et al., 2021). these modeling data can expect the active sites and the possible reaction mechanism in biohydrogen production.
- 6.6 Unstable performances and relatively high ratio of energy (or capital) input to energy output might also remain challenge.
- 6.7 MEC has not yet achieved practical application in a wide range, which is mainly tested at lab-scale. Moreover, some other individual technologies of this hybrid system (e.g., phosphate removal based-high rate activated sludge process) are also not applied in real-world. Hence, scaling up this hybrid system would be therefore inevitable.
- 6.8 Obstacles of the potential of hybridization of MEC with other processes in full-scale application (e.g. long distance between electrodes and pH gradient would lead to high internal resistances.) Besides, high capital costs of expensive electrodes and membrane materials also limit its application. Also, there are still lacking comprehensive understanding of microbiology involved in microbial electrolysis cells including fermentative bacteria, exoelectrogens and biocathode microorganisms. By further designing microbial electrolysis cell reactors, developing new materials, deeply understanding and taking control over microbiology are expected to address these challenges. it's crucial to acquire government support to develop such emerging technologies (Fu et al., 2021).
- 6.9 Focus on the entire lifecycle stages not on specific parts of the hydrogen cycle (e.g. production, storage) (Dawood et al., 2020).

7. Conclusions

This review highlights the potential of biohydrogen as a future clean and green energy source that can contribute significantly to the reduction of fossil fuel consumption. AnMBR is a promising technology to achieve high-rate hydrogen production, but its adoption at industrial scale is hindered by the high capital and operation cost. However, the decreasing price of membranes, and the development of innovative solutions such as dynamic membrane reactors, can promote scale-up of this technology. Membrane fouling remains the main challenge to be addressed, particularly when treating wastewaters with high solid content. ED can be combined to dark fermentation for in-line extraction of VFAs, mitigating product inhibition. Combination of ED with liquid-liquid separation is a potential solution to achieve selective VFA extraction while maintaining in the reactor ions and nutrients required for microbial growth. MEC is still an immature technology, which at present cannot compete with dark fermentation in terms of costs and hydrogen production rate. However, MEC will benefit of the decreasing cost trend of renewable electricity, which can make it competitive if technological development brings cheaper construction materials and higher H₂ production rates.

Credit author statement

Mohamed El-Qelish: Conceptualization, Investigation, Writing most

of the original draft, Reviewing, and Editing, Gamal K. Hassan: Conceptualization, Investigation, Writing most of the original draft, Reviewing, and Editing, Sebastian Leaper: Reviewing and Editing, Paolo Dessì: Writing original draft, Reviewing and Editing, Ahmed Abdel-Karim: Investigation, Writing original draft, Reviewing, and Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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